

Statistical modeling

Module 08

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Bachelor, GSEM

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Regression for research

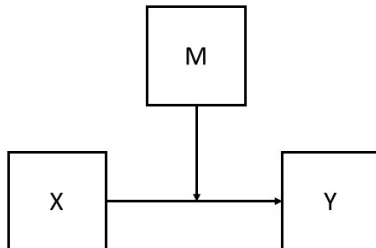
In research, most common questions regarding the study of a response y compare to variables x are:

- ▶ Does a given variable x_j moderates the effect of another variable x_k ? If x_j is large, does the effect on y of x_k different from the one when x_j is small? (Moderation)
- ▶ Does a given factor mediates the effect of another factor? Is the effect of x_j on y direct or just the consequence of its effect on x_k ? (Mediation)
- ▶ Does a given variable x_j has an impact? When x_j varies, does y varies accordingly? (Association and Causality)

Regression for research: Moderation

Moderation is a synonym of interaction. It is often of primary interest in psychology, social science, marketing, etc.

The graphical representation of a moderation effect is



The effect of X on Y is modified when M changes.

Moderation

Example: Burn out case: "Is the effect of socialization on burn out the same for both sexes?"

Compare the models with and without the interactions:

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~ Social+I(Social^2) + Sex, data=ashley)
> anova(fit0, fit1)
```

Analysis of Variance Table

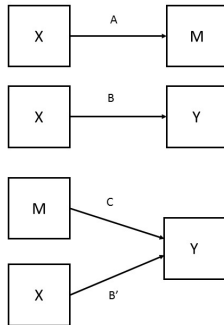
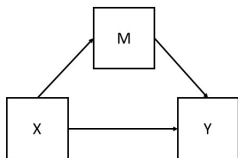
```
Model 1: BurnOut ~ Social + I(Social^2) + Sex
Model 2: BurnOut ~ (Social + I(Social^2)) * Sex
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	16	231845				
2	14	127317	2	104528	5.747	0.01506 *

Answer: yes, factor Sex is a moderator of factor Social (at a significance level of 5%).

Regression for research: Mediation

Mediation means that one factor has its effect partially or fully through the other factor, the mediator. It is a composite hypothesis (i.e. it includes several hypotheses).



Mediation

Mediation means:

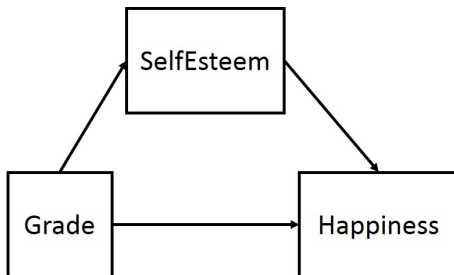
- ▶ Effect A is significant: X has an effect on M .
- ▶ Effect B is significant: X has an effect on Y .
- ▶ Effect B' is lower than B in absolute value: knowing M , the effect of X on Y is smaller (or null).

With mediation, we can conclude that the X has an effect on M which in turn has an effect on Y , or that the effect of X on Y is through M .

Mediation: example

Does self-esteem mediate the effect of grades on happiness?

adapted from <http://data.library.virginia.edu/introduction-to-mediation-analysis/>



Mediation: direct and indirect effects

Indirect and direct effects are often of interest.

If there is a mediation, then the total effect of X on Y is

- ▶ B in the regression $X \rightarrow Y$.
- ▶ $B' + A \times C$ in regression the $X + M \rightarrow Y$.

Both should be equal:

$$B = B' + A \times C.$$

Also

- ▶ $A \times C$ is called the indirect effect of X on Y through M .
- ▶ B' is the direct effect.
- ▶ B is the total effect.

Mediation: example

Three regressions:

```
> summary(fitXY <- lm(Happiness ~ Grades, data=myData))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.8572	0.6932	4.122	7.88e-05 ***
Grades	0.3961	0.1112	3.564	0.000567 ***

```
[...]
```

```
> summary(fitXM <- lm(SelfEsteem ~ Grades, data=myData))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.49952	0.58920	2.545	0.0125 *
Grades	0.56102	0.09448	5.938	4.39e-08 ***

```
[...]
```

```
> summary(fitXMY <- lm(Happiness ~ Grades+SelfEsteem, data=myData))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.9043	0.6055	3.145	0.0022 **
Grades	0.0396	0.1096	0.361	0.7187
SelfEsteem	0.6355	0.1005	6.321	7.92e-09 ***

```
[...]
```

Grades depends on happiness, self-esteem depends on grade, the effect of grades vanishes when self-esteem is considered: it's a full mediation.

Note: to test the significance of the mediation, in fact specific tests should be used.

Mediation: example

In the example,

$$B = 0.3961 = 0.0396 + 0.56102 \times 0.6355 = B' + A \times C$$

- ▶ The total effect is 0.3961
- ▶ The indirect effect is $0.56102 \times 0.6355 = 0.3565$
- ▶ The direct effect is 0.0396.

Considering the non-significance, one could say that the direct effect is 0, and run a regression without Grade

SelfEsteem $\longrightarrow$ Happiness

to compute C and the indirect effect.

Association

Association is not Causality. These two concepts should not be confounded!!

Two variables (or factors) are associated if they are observed varying together.

Correlation (Pearson, Spearman, Kendall) is a measure of association between 2 variables.

Regression helps also describing the association between the response y and a set of variables x . In this context, R^2 's and F -value are measures of association. The coefficients and the p -values are conditional measures of association in the sense that they are estimated given the other variables.

Association and causality

Example: With the burn out data, the researchers want to know if "the socialization index and the burn out index are associated"

From the quadratic model with interaction (`fit1`), the question can be answered by testing the effect of factor `Social` on `BurnOut`. Below with an F -test:

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~Sex, data=ashley)
> anova(fit0, fit1)
```

Analysis of Variance Table

Model 1: `BurnOut ~ Sex`

Model 2: `BurnOut ~ (Social + I(Social^2)) * Sex`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	18	1199408				
2	14	127317	4	1072090	29.472	1.102e-06 ***

Answer: yes, `BurnOut` and `Social` are associated (at a significance level of 5%).

Association and causality

In previous example, it **cannot** be concluded that factor Socialization **causes** the Burn Out index. We can only conclude that these two factors are **associated**: they have been observed varying together.

Without being too philosophical, causality is a more profound concept that implies a hierarchy into the causes:

x causes y

means that

*a change in x keeping all the other factors constant implies
(causes) a change in y*

Association and causality

Example: in the burn out study, the explanation could be

- ▶ The Burn Out index causes the Socialization: subject with a large burn out probability will little by little de-socialize themselves causing the social index to move.

OR

- ▶ Another unobserved factor causes both Burn Out index and Socialization index. For example, it could be an economical index. A small economical index would imply low job quality, low salary, (etc.) causing both burn out favorable field and de-socialization.

Association and causality

Can we really assess/measure causality? It relies on

- ▶ A design of experiments: observational study are often limited by the randomness of the sampling whereas planned experiments offer a better control on the effect of the factors.
- ▶ A scientific analysis: the answer to the question "what causes what" is often the result of a long process of knowledge acquisition, both in terms of understanding the links, and in terms of which meaningful factors should be included in the study.

In the understanding of causation, statistical analysis is just one tool among others.

Association and causality

Examples: In the burn out study,

- ▶ the fact that burn out causes socialization or the other way cannot be distinguished without a specific analysis and logical reflexion by scientists of the domain. The regression (or any statistical analysis) cannot help here.
- ▶ The fact that there may exist another factor which drives both burn out and socialization indexes cannot be known without an expertise of a scientist in this domain. His/her own knowledge should result from a large number of other studies in this field.

The burn out study is observational. It is a classical limit of observational studies.

Association and causality

Examples (continued): A yield of a production process of some chemical product Y is studied with respect to the quantity of incoming products A and B and the temperature of the cooking chamber. All these factors are perfectly controlled and a plan of experiments is made where several properly chosen combinations of the factors are tested. At the end, it is discovered that the yield increases with the temperature. There, it can be concluded (under the guidance of a chemist) that the temperature causes the yield (in a certain range). Indeed,

- ▶ Since all the factors are perfectly controlled, it is known that their level is not the result of another unobserved factor.
- ▶ Since the temperature is perfectly controlled, it is not the yield which could cause it.

It is an example of a planned study. When they are available, they provide stronger conclusions than observational ones.

Analysis of collinearity

One of the common problems in observational studies is the collinearity of the factors.

Collinearity means that two (or more) factors can vary together so that their effect on a response variable in a regression cannot be distinguished.

This can be detected when there exist correlations (or more generally dependence) between these factors.

Example: The cars example `?mtcars`. The data was extracted from the 1974 Motor Trend US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973–74 models).

Analysis of collinearity

A data frame with 32 observations on 11 variables.

- ▶ mpg: Miles/(US) gallon - the response
- ▶ cyl: Number of cylinders (4-6-8)
- ▶ disp: Displacement (cu.in.)
- ▶ hp: Gross horsepower
- ▶ drat: Rear axle ratio
- ▶ wt: Weight (1000 lbs)
- ▶ qsec: 1/4 mile time
- ▶ vs: V-engine/Straight engine (0/1)
- ▶ am: Transmission (0 = automatic, 1 = manual)
- ▶ gear: Number of forward gears
- ▶ carb: Number of carburetors

Analysis of collinearity

```
> summary(lm(mpg~., data=mtcars))
```

Call:

```
lm(formula = mpg ~ ., data = mtcars)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.4506	-1.6044	-0.1196	1.2193	4.6271

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.30337	18.71788	0.657	0.5181
cyl	-0.11144	1.04502	-0.107	0.9161
displ	0.01334	0.01786	0.747	0.4635
hp	-0.02148	0.02177	-0.987	0.3350
drat	0.78711	1.63537	0.481	0.6353
wt	-3.71530	1.89441	-1.961	0.0633
qsec	0.82104	0.73084	1.123	0.2739
vs	0.31776	2.10451	0.151	0.8814
am	2.52023	2.05665	1.225	0.2340
gear	0.65541	1.49326	0.439	0.6652
carb	-0.19942	0.82875	-0.241	0.8122

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.65 on 21 degrees of freedom

Multiple R-squared: 0.869, Adjusted R-squared: 0.8066

F-statistic: 13.93 on 10 and 21 DF, p-value: 3.793e-07

Analysis of collinearity

In particular, it means that `cyl` may be predictable with a linear model containing all the other variables (except `mpg`).

If this is true, it means that, approximately,

$$\text{cyl} \approx a_0 + a_1 \text{disp} + \cdots + a_9 \text{carb}$$

If this is true, then `cyl` is useless in the regression on `mpg`

$$\text{mpg} = b_0 + b_1 \text{cyl} + b_2 \text{disp} + \cdots + b_{10} \text{carb} + e$$

which it can be replaced by

$$\text{mpg} = c_0 + c_1 \text{disp} + \cdots + c_9 \text{carb} + e$$

for some $c_0 = b_0 + a_0$, $c_1 = b_2 + c_1 \times a_1$, etc.

In particular, a test on the coefficient b_1 will be non-significant. This is in fact exactly what happened...

Analysis of collinearity

If a research is based on that model (i.e. with collinearities), then the scientific conclusions could be incorrect: `tt` `cyl` and `mpg` are not associated when they are.

"This is so horrible, I can't live with this! How can we avoid that?"

- ▶ Once collinearity is here, this confusion in the conclusion cannot be avoided.
- ▶ Avoiding collinearity can only be done using a design of experiments.
- ▶ At least, we should be informed about collinearity to modulate our conclusions

To analyse collinearity, use the Variance Inflation Factor (VIF).

VIF

The Variance Inflation Factor (VIF) quantifies how much $s_{\hat{b}_j}$, the standard error of the coefficient estimate $\hat{b}_j$, is inflated when multicollinearity exists.

It is computed from the regression for each variable x_j as

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 of the regression of x_j on all the other factors (i.e. x_{-j}).

The interpretation of the VIF is the inflation of the variance of $\hat{b}_j$ due to the collinearity of x_j with the other variables¹. From the previous formulas,

1. If x_j is independent of the other variables then $R_j^2 \approx 0$, and $VIF_j \approx 1$.
2. If x_j is very collinear to the other variables then $R_j^2 \approx 1$, and $VIF_j \approx \infty$.

A rule of thumb commonly admitted is

- ▶ a VIF of 1 means no collinearity.
- ▶ a VIF greater than 4 raises a warning for further investigations.
- ▶ a VIF greater than 10 is a severe sign of collinearity.

¹thus the name, VIF.

Analysis of collinearity: VIF

Example:

```
> fit <- lm(mpg~., data=mtcars)
> vif(fit)
```

cyl	disp	hp	drat	wt	qsec	vs
15.373833	21.620241	9.832037	3.374620	15.164887	7.527958	4.965873
am	gear	carb				
4.648487	5.357452	7.908747				

Thus, collinearity is detected on `cyl`, `disp`, and `wt`. Conclusions from tests on these variables in a research context should be taken with care.

Analysis of collinearity and variable selection

When there is no collinearity, a non-significant variable can be removed from the model. This is called variable selection or model selection. The aim is to build a model which is as simple as possible, as complex as needed.

If there are collinearities, say between x_1 and x_2 , then it should always be kept in mind that removing x_1 from the model is (almost) equivalent to removing x_2 . This is purely arbitrary.

Variable selection: why is it important?

In research,

- ▶ Simplify the model to strengthen the conclusion and provide interpretability (i.e. with lots of variables, conclusion can become very complex to draw).
- ▶ Check whether the sample is representative of a population (e.g. if a variable is known from previous studies but is not selected in the new sample).

In prediction,

- ▶ Get interpretable models,
- ▶ Limit the risk of missing values,
- ▶ Control resources (maintain/track variables can be very costly).

Variable selection

Example: car consumption explained by all the variables.

```
> fit <- lm(mpg ~ ., data=mtcars)
> summary(fit)
```

Call:

```
lm(formula = mpg ~ ., data = mtcars)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.4506	-1.6044	-0.1196	1.2193	4.6271

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
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displacement	0.01334	0.01786	0.747	0.4635
hp	-0.02148	0.02177	-0.987	0.3350
drat	0.78711	1.63537	0.481	0.6353
wt	-3.71530	1.89441	-1.961	0.0633
qsec	0.82104	0.73084	1.123	0.2739
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.65 on 21 degrees of freedom

Multiple R-squared: 0.869, Adjusted R-squared: 0.8066

F-statistic: 13.93 on 10 and 21 DF, p-value: 3.793e-07

Variable selection using tests

Individual t-tests show that each variable is not significant **when keeping the others**. However, the model is significant as shown by the F -test. It means that some variables are significant.

Which variables should be kept in the model?

Two approaches:

- ▶ Stepwise selection: remove non-significant variables (at level α) one at a time.
- ▶ Information criterion: select the model with minimum criterion.

Variable selection using tests

In the previous example, for $\alpha = 5\%$, cyl has the highest p -value, we can remove it:

```
> summary(lm(mpg ~ .-cyl, data=mtcars))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.96007	13.53030	0.810	0.4266
displacement	0.01283	0.01682	0.763	0.4538
horsepower	-0.02191	0.02091	-1.048	0.3062
drat	0.83520	1.53625	0.544	0.5921
weight	-3.69251	1.83954	-2.007	0.0572
qsec	0.84244	0.68678	1.227	0.2329
vs	0.38975	1.94800	0.200	0.8433
am	2.57743	1.94035	1.328	0.1977
gear	0.71155	1.36562	0.521	0.6075
carb	-0.21958	0.78856	-0.278	0.7833

Now remove vs, then repeat again and again... Final model:

```
> summary(lm(mpg ~ .-cyl-vs-carb-gear-drat-disp-hp, data=mtcars))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.6178	6.9596	1.382	0.177915
weight	-3.9165	0.7112	-5.507	6.95e-06 ***
qsec	1.2259	0.2887	4.247	0.000216 ***
am	2.9358	1.4109	2.081	0.046716 *

No more effect can be removed with a test on the coefficients.

Variable selection using tests

The procedure is one of the most used, although it is heuristic.

To be sure, one should do a additional check with a F -test comparing the full model and the simplified one:

```
> fit <- lm(mpg ~ ., data=mtcars)
> fit0 <- lm(mpg ~ .-cyl-vs-carb-gear-drat-disp-hp, data=mtcars)
> anova(fit, fit0)
[...]
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	147.49				
2	28	169.29	-7	-21.791	0.4432	0.8636

The final selection is thus validated: the selected model is (statistically) identical to the most complex one.

Variable selection using tests

Recommendations and comments:

- ▶ Stepwise selection can be used for model growing (adding variables instead of removing them).
- ▶ The rule "remove the highest p-values" is heuristic.
- ▶ At each step any non-significant variable is a candidate for removal, several final models are possible.
- ▶ A scientific prior can also guide into the variable selection: e.g. keep one variable of interest as long as possible.

Selection with tests is an old methods. More and more alternatives are now possible.

Variable selection based on information criterion

Variable selection can be based on an information criterion. The principle is the law of parsimony or Occam's razor:

"Among competing models, the one with the fewest assumptions should be selected"

For statistical models this translates into

"Between models fitting equally well to the data, prefer the simplest one."

A good model should thus achieve a trade-off between goodness-of-fit and simplicity. The AIC measures this tradeoff.

Variable selection based on information criterion

The Akaike Information Criterion (AIC) of a model M is

$$AIC(M) = -2\ell_M + 2K$$

where

- ▶ K is the total number of parameters in model M ,
- ▶ ℓ_M is the maximum log-likelihood of model M .

Variable selection based on information criterion

For regression,

- ▶ $K = p + 2$: intercept, one coefficient per variable (incl. dummy variables), σ .
- ▶ The maximum log-likelihood satisfies

$$-2\ell_M = n \log 2\pi - n \log n + n + n \log SS_e$$

- ▶ The AIC is thus

$$AIC \equiv n \log SS_e - 2(p + 2).$$

It indeed measures a compromise between model complexity ($p + 2$) and model error (SS_e).

The $AIC(M)$ should be as small as possible.

Variable selection based on information criterion

Automatic: At each step, remove the variable for which AIC decreases the most. Stop when no removal decreases AIC anymore.

```
> step(fit, direction="backward")
Start:  AIC=70.9
mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb
```

	Df	Sum of Sq	RSS	AIC
- cyl	1	0.0799	147.57	68.915
- vs	1	0.1601	147.66	68.932
- carb	1	0.4067	147.90	68.986
- gear	1	1.3531	148.85	69.190
- drat	1	1.6270	149.12	69.249
- disp	1	3.9167	151.41	69.736
- hp	1	6.8399	154.33	70.348
- qsec	1	8.8641	156.36	70.765
<none>			147.49	70.898
- am	1	10.5467	158.04	71.108
- wt	1	27.0144	174.51	74.280

```
[...]
Step:  AIC=61.31
mpg ~ wt + qsec + am
```

	Df	Sum of Sq	RSS	AIC
<none>			169.29	61.307
- am	1	26.178	195.46	63.908
- qsec	1	109.034	278.32	75.217
- wt	1	183.347	352.63	82.790

```
[stop]
```

Variable selection based on information criterion

Other information criteria exist:

- ▶ Bayesian Information Criteria: $BIC = -2\ell_M + K \log n$
- ▶ $AIC_c = AIC + 2K(K + 1)/(n - K - 1)$
- ▶ etc.

Different searches can be performed:

- ▶ Backward: see before;
- ▶ Forward: add one variable at each step;
- ▶ Both: try add or remove at each step.

Variable selection: discussion

When to use which method?

- ▶ In general, stepwise selection with tests is preferred in a research context: the final aim is a test. It is the method which allows a good control on the model selection and gives the highest chance to finish with a meaningful model.
- ▶ In a prediction context (e.g. machine learning, business analytics), AIC (and friends) is preferred. The interpretation of the model is less important, only its prediction capacity and its simplicity are.
- ▶ It is of course possible to remove significant variables with AIC or to have non-significant variable at the end of a AIC selection. These two procedures are just different.
- ▶ Test based or AIC selection can be preceded by a one-by-one variable selection based on the VIF.